

Data Structures and Algorithms · 21CSC201J

Unit 1 · Session 7 · SLO 1 — Solved Worksheet

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Unit “ 1 : Session “ 7 : SLO “ 1 | Data Structures “ Definition and Types

■ I. Fill in the Blanks

1. data structure “ a way of organizing and storing data for efficient access and modification.
2. linear and non-linear “ the two broad types.
3. linear “ elements arranged sequentially.
4. Queue “ the FIFO linear data structure.
5. Tree “ the non-linear structure for hierarchical relationships.
6. graph “ elements point to multiple elements, forming a network.
7. push and pop “ the two stack operations.
8. array “ a collection of elements identified by index.
9. linked list “ nodes containing data plus a next pointer.
10. storage (organization) and retrieval (processing) “ efficient management of data.

■ II. Match the Following

1. Array “ C. Sequential memory allocation.
2. Stack “ A. Last In First Out.
3. Queue “ B. First In First Out.
4. Linked List “ D. Each node points to the next node.
5. Tree “ J. Hierarchical structure.
6. Graph “ H. Represents complex relationships.
7. Linear Data Structure “ F. Stack (example).
8. Non-Linear Data Structure “ I. Graph (example).
9. Stack Operations “ G. Push and Pop.
10. Queue Operations “ E. Enqueue and Dequeue.